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Expt No 1 Basic Networking Commands

Aim:

Study of various network commands used in linux and windows.

Commands:

(i) arp - a :

ARP is a short form of address resolution protocol which shows the IP address of your command along with IP address and MAC address of your router.

Output :

```
C:\Users\sri sadhararp -a

Interface: 169.254.132.119 --- 0xb
  Internet Address      Physical Address      Type
  169.254.255.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 169.254.192.205 --- 0xc
  Internet Address      Physical Address      Type
  169.254.255.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 10.98.125.149 --- 0x14
  Internet Address      Physical Address      Type
  10.98.125.211         9e-37-67-6c-3b-79     dynamic
  10.98.125.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

(ii) hostname

It displays the name of your computer.

Sample output :

RUPA

(iii) ipconfig/all :

This command displays detailed configured information about your TCP/IP connection.

Output:

```
C:\Users\Ari sadhu>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-35NLPQ9L
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Mixed
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0E
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::80cf:8e7c:d90d:6fda%12(Preferred)
   Autoconfiguration IPv4 Address. . : 169.254.192.205(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 687301583
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-D9-65-D4-94-E0-61-18-C3-25
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet 3:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter #2
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0A
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::79f6:1a31:6a50:aa81%11(Preferred)
   Autoconfiguration IPv4 Address. . : 169.254.112.114(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 620521255
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2F-D9-65-D4-94-E0-61-18-C3-25
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 96-E0-43-18-C3-25
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

(iv) hbtstat - a:

This command helps solve problems with NetBIOS name resolution.

Output:

```
C:\Users\Ari sadhu>nbtstat -a

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP)

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -s   (Sessions)       Lists sessions table with the destination IP addresses
  -S   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

(v) netstat

netstat displays a variety of statistics about a computer active TCP/IP connections. Its command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables and interfaces.

```
C:\Users\sri sadha>netstat -r
===========================================================================
Interface List
 12...0a 00 27 00 00 0c ......VirtualBox Host-Only Ethernet Adapter
 17...0a 00 27 00 00 0b ......VirtualBox Host-Only Ethernet Adapter #2
 13...9e bb 43 1b c3 25 ......Microsoft Wi-Fi Direct Virtual Adapter #3
 18...9e bb 43 1b c3 35 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 26...94 bb a3 3b c3 25 ......MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    10.90.125.211    10.90.125.149     50
       10.90.125.0    255.255.255.0         On-link     10.90.125.149    306
     10.90.125.149  255.255.255.255         On-link     10.90.125.149    306
     10.90.125.255  255.255.255.255         On-link     10.90.125.149    306
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      169.254.0.0      255.255.0.0         On-link   169.254.192.205    281
      169.254.0.0      255.255.0.0         On-link   169.254.132.119    281
  169.254.132.119  255.255.255.255         On-link   169.254.132.119    281
  169.254.192.205  255.255.255.255         On-link   169.254.192.205    281
  169.254.255.255  255.255.255.255         On-link   169.254.192.205    281
  169.254.255.255  255.255.255.255         On-link   169.254.132.119    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link   169.254.192.205    281
        224.0.0.0        240.0.0.0         On-link   169.254.132.119    281
        224.0.0.0        240.0.0.0         On-link     10.90.125.149    306
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link   169.254.192.205    281
  255.255.255.255  255.255.255.255         On-link   169.254.132.119    281
  255.255.255.255  255.255.255.255         On-link     10.90.125.149    306
===========================================================================
Persistent Routes:
  None
```

(vi) nslookup

It is used to perform DNS lookups in linux.

Ex. nslookup www.google.com

```
C:\Users\sri sadha>nslookup www.google.com
Server:  UnKnown
Address:  10.90.125.211

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482b:7700::
          2001:4860:4829:7700::
          2001:4860:482c:7700::
          2001:4860:4827:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4828:7700::
          142.251.153.119
          142.251.152.119
          142.251.157.119
          142.251.151.119
          142.251.154.119
          142.251.150.119
          142.251.156.119
          142.251.155.119
```

(vii) pathping

pathping is unique to windows and is basically a combination of the ping and traces command.

```
C:\Users\sri sadha>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

(viii) ping - Command is the best way to test the connection between two computers.

```
C:\Users\sri sadha>ping LAPTOP-5SNLPGRL

Pinging LAPTOP-5SNLPGRL [fe80::38f3:7a6c:ae35:d00d%20] with 32 bytes of data:
Reply from fe80::38f3:7a6c:ae35:d00d%20: time<1ms
Reply from fe80::38f3:7a6c:ae35:d00d%20: time<1ms
Reply from fe80::38f3:7a6c:ae35:d00d%20: time<1ms
Reply from fe80::38f3:7a6c:ae35:d00d%20: time<1ms

Ping statistics for fe80::38f3:7a6c:ae35:d00d%20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

(ix) Route:

Route command is used to show / manipulate the IP routing table.

```
C:\Users\sri sadha>route print
===========================================================================
Interface List
 12...0a 00 27 00 00 0c ......VirtualBox Host-Only Ethernet Adapter
 11...0a 00 27 00 00 0b ......VirtualBox Host-Only Ethernet Adapter #2
 13...96 bb a9 1b c3 23 ......Microsoft Wi-Fi Direct Virtual Adapter #3
 18...96 bb a9 1b c3 33 ......Microsoft Wi-Fi Direct Virtual Adapter #4
 20...94 9a a8 1b c3 25 ......MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    10.90.125.211    10.90.125.149     50
       10.90.125.0    255.255.255.0         On-link     10.90.125.149    306
     10.90.125.149  255.255.255.255         On-link     10.90.125.149    306
     10.90.125.255  255.255.255.255         On-link     10.90.125.149    306
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      169.254.0.0      255.255.0.0         On-link    169.254.192.205    281
    169.254.192.0    255.255.255.0         On-link    169.254.192.205    281
    169.254.192.205  255.255.255.255       On-link    169.254.192.205    281
    169.254.232.0    255.255.255.0         On-link    169.254.232.119    281
    169.254.232.119  255.255.255.255       On-link    169.254.232.119    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link    169.254.192.205    281
        224.0.0.0        240.0.0.0         On-link    169.254.232.119    281
        224.0.0.0        240.0.0.0         On-link     10.90.125.149    306
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link    169.254.192.205    281
  255.255.255.255  255.255.255.255         On-link    169.254.232.119    281
  255.255.255.255  255.255.255.255         On-link     10.90.125.149    306
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
 20     60 ::/0                     fe80::9e37:a7ff:febc:3b70
  1    331 ::1/128                  On-link
 20    306 2409:40f4:40ec:7d23::/64 On-link
 20    306 2409:40f4:40ec:7d23:1c11:7c47:c68:e506/128
                                    On-link
 20    306 2409:40f4:40ec:7d23:95ff:8bdd:4e93:bd8e/128
                                    On-link
 12    281 fe80::/64                On-link
 15    281 fe80::/64                On-link
```

Important Linux Networking Commands

1. ip: ip command is one of the basic commands every administrator will need in daily work from entry up new system.

2. ifconfig: The ifconfig command was a staple in many sysadmin tool belt for configuring and troubleshooting networks. It has now been replaced by the ip commands discussed above.

3. mtr: Serves as a network diagnostic tool.
mtr -g google.com

4. tcp dump: The tcp command designed for capturing and displaying packets.
tcpdump -i

5. ping: Tool to verify if local connectivity to another TCP/IP computer by sending internet control message protocol echo request messages.
ping localhost.

8/12/2020

Result:

The various network commands given in the exercise were studied and executed successfully.

Exp No 2 Study of Network Cables

Aim:
To study different types of network cables.

(a) Understand different types of network cables

Cable Type	Category	Max. data transfer rate	Advantages & Disadvantages	Application
(i) UTP	3	10 Mbps	(i) cheaper (ii) More prone to EMI	10-base T-ethernet
(ii) STP	6, 6a	10 Gbps	(i) shielded (ii) Expensive	used in data centers
(iii) SSTP	7	10 Gbps	(i) Greater effort install	10 G ethernet
(iv) Coaxial cable	RG-6 RG-59	10-100 mbps	(i) High bandwidth (ii) size is bulky	TV network
(v) Fibre Optics cable	Single mode	100 Gbps	(i) High speed (ii) Expensive	Max distance is 100 m

b) Make your own Ethernet cross over cable

Step-1: To start construction of device, begin with already shield onto cable.

Step-2: Next, strip approximately 1.5 cm of cable shielding from both ends.

Step-3: After you need to untangle the wire three should be 4 twisted pairs.

Step-4: Once the order is correct, bunch them together in line, and if there are any that stick out farther than other, snip them back to create an even level node.

Step-5: Next push the cable right in: The node at the end of the plug needs to jump over the cable shielding.

Step-6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step-7: Lastly, repeat for other end.

Result:

The different types of network cables is studied.

Expt. No. 3 CISCO Packet TRACER

Aim:

To study the packet tracer tool installation and user interface.

a) UI overview:

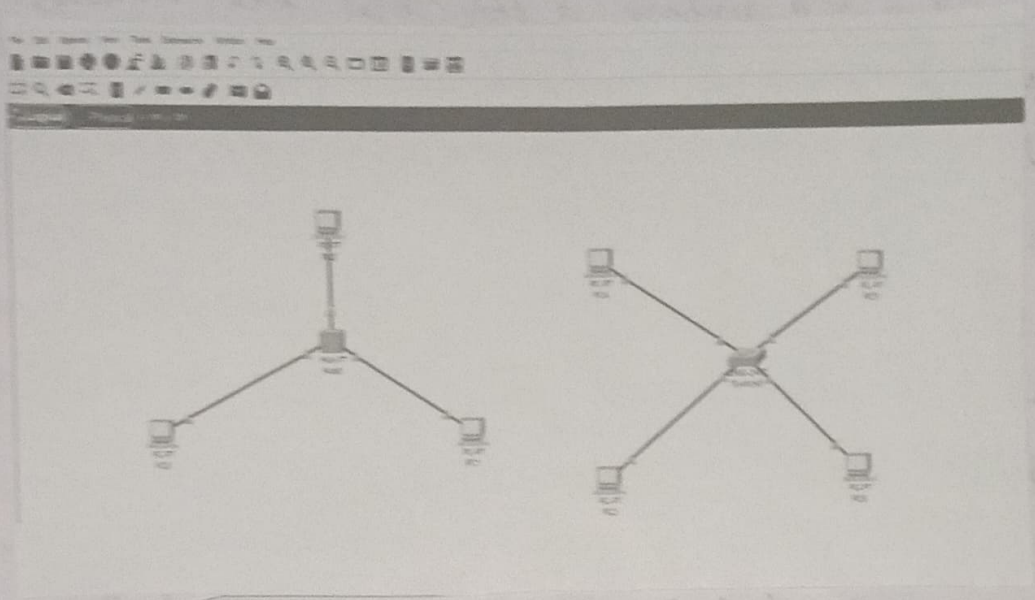
The layout of packet tracer is divided into several components.

The components are:

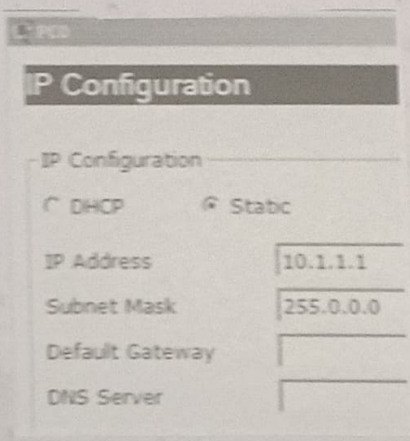
(i) Menu bar
(ii) Menu toolbar
(iii) Logical tabs.
(iv) Workspace
(v) Real time tabs
(vi) Network component box
(vii) Common tool box
(viii) User created environment.

b) Analyse behavior of network devices using CISCO Packet Tracer simulation.

1) From network box, drag and drop the components
2) click on connection
   a) click on copper straight through cable.
   b) connect 4 PCs to switch.
3) Click on PC connected to hub, go to desktop tab, click on IP configuration and enter IP address and subnet mask. Here the default gateway and DNS server information is not needed as there are only two end devices connected in a network.



4) Observe the flow of PDU from source PC to destination PC by selecting realtime mode.



5) Repeat steps 3 to 4.

6) Observe how hub and switch are forwarding the PDU and write your observation.

Result:

The CISCO packet tracker [TRACER] is successfully studied.

Expt No. 4 Setup and configure LAN

Setup and configure a LAN (Local Area Network) using a switch and ethernet cables in your lab

How to setup a LAN:

Step-1: Plan and design an appropriate network topology taking its account network requirements and equipment location.

Step-2: You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes and 4 ethernet cables.

Step-3: Connect your computer to network switch via an ethernet cable which is as simple as plugging one end of the ethernet cable into your computer and the other end into your network switch.

Step-4: Assign IP addresses to your PCs.

Step-5: Configure a network switch

1. Connect your computer to the switch: To access the switch's web interface, you will need to connect your computer to the switch using an ethernet cable.

2. Log in to the web interface: Open a web browser and enter the IP address of the switch in the address bar. This should bring up the login page for the switch's web interface. Enter the username and password to log in.

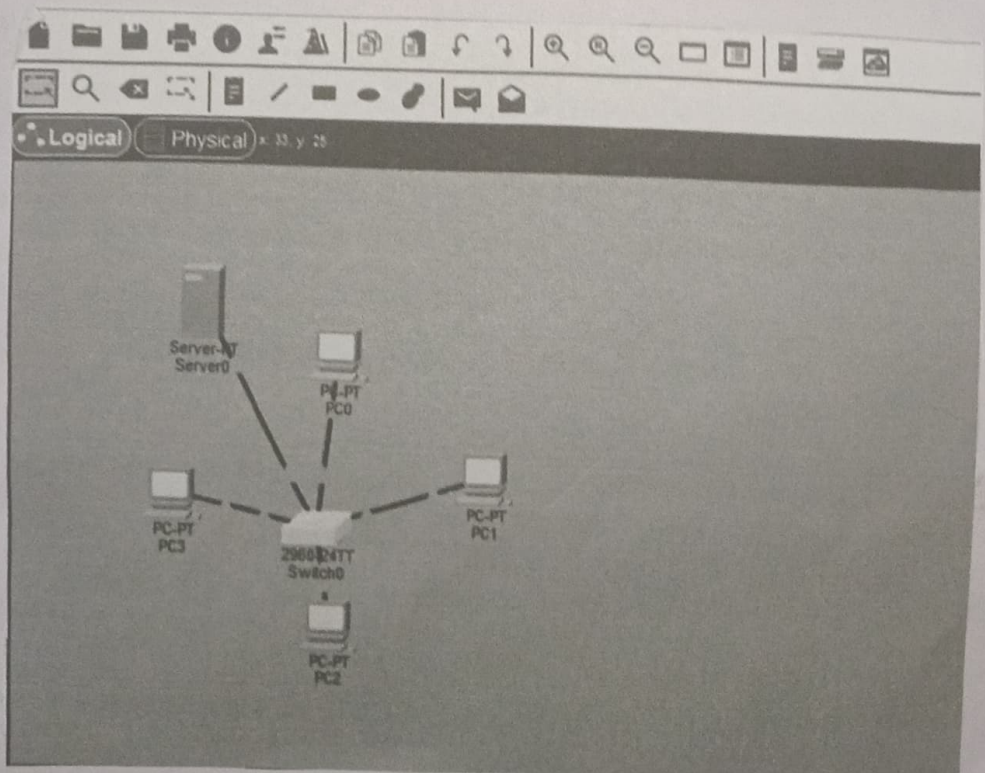
3. Configure basic settings. Once you logged in, you will be able to configure basic settings for the switch.

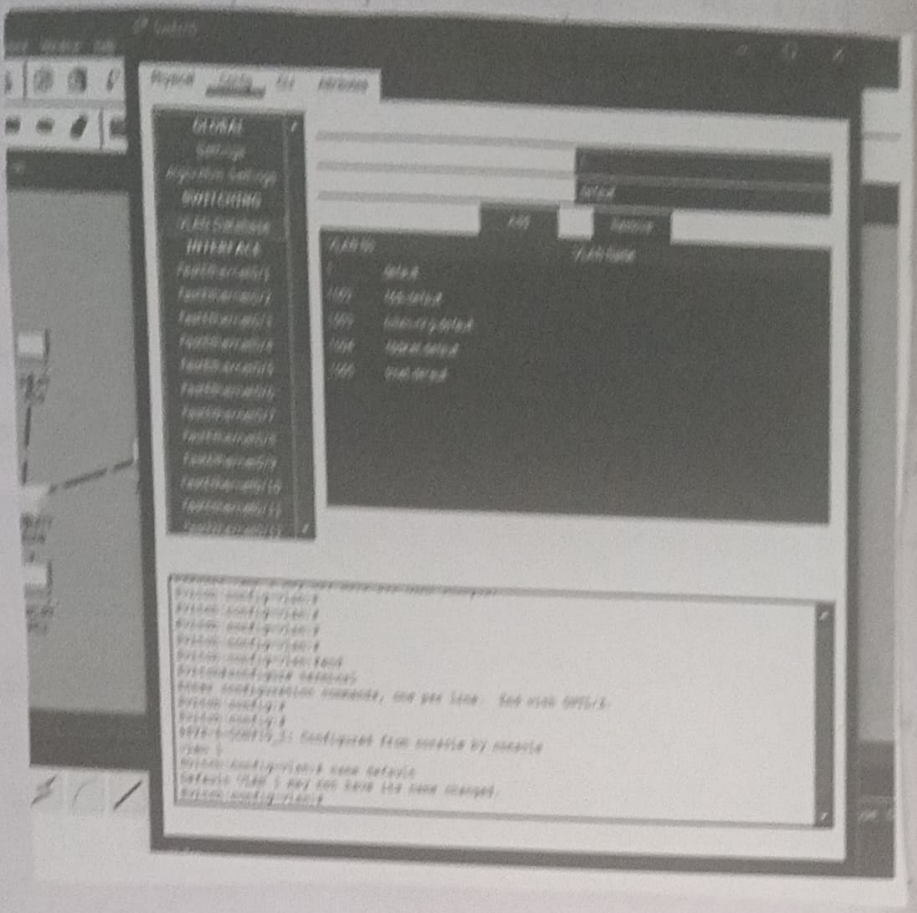
4. Assign IP address as 10.1.1.5 ; subnet mask 255.0.0.0

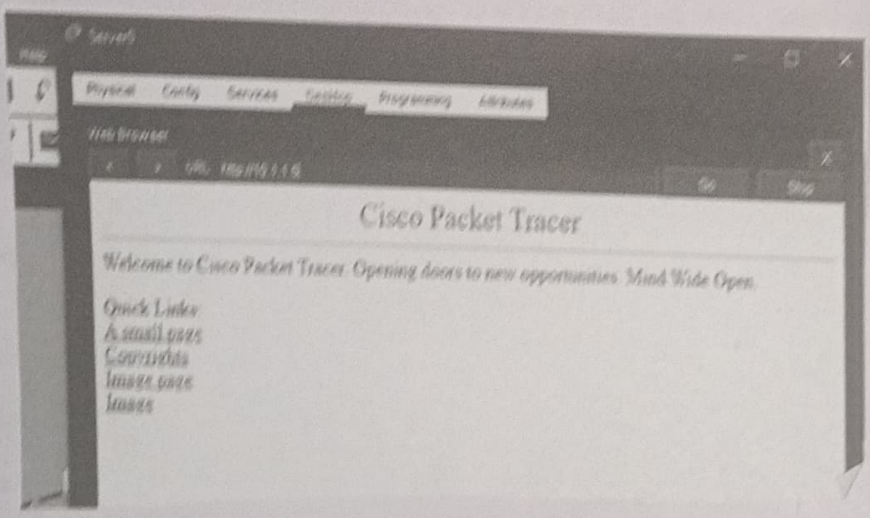
5. Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

6. Select a folder -> Go to properties -> Click sharing Tab -> share it with everyone on the same LAN.

7. Try to access the shared folder from other computers of the network.







Result:

The LAN is configured and set up successfully.

2/3/23

Expt. No. 5 Packet capturing tool - Wireshark

Aim:

Experiments on packet capture tool : Wireshark packet sniffer.

-> Sniffs messages being sent/received from/by your computer.

-> Store and display the contents of various protocol fields in the message.

Procedure:

-> Select local area connections in Wireshark

-> Go to capture option

-> Then select stop capture automatically after 100 packets.

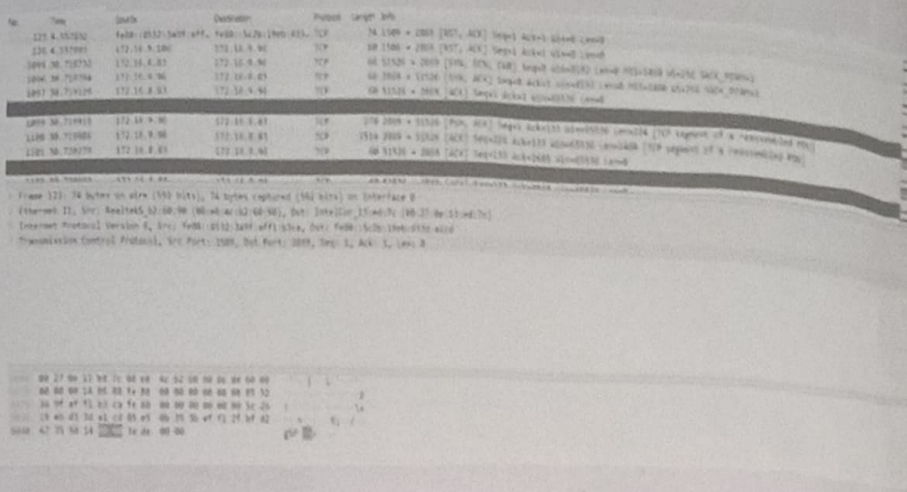
-> Then click start capture.

-> Save the packets.

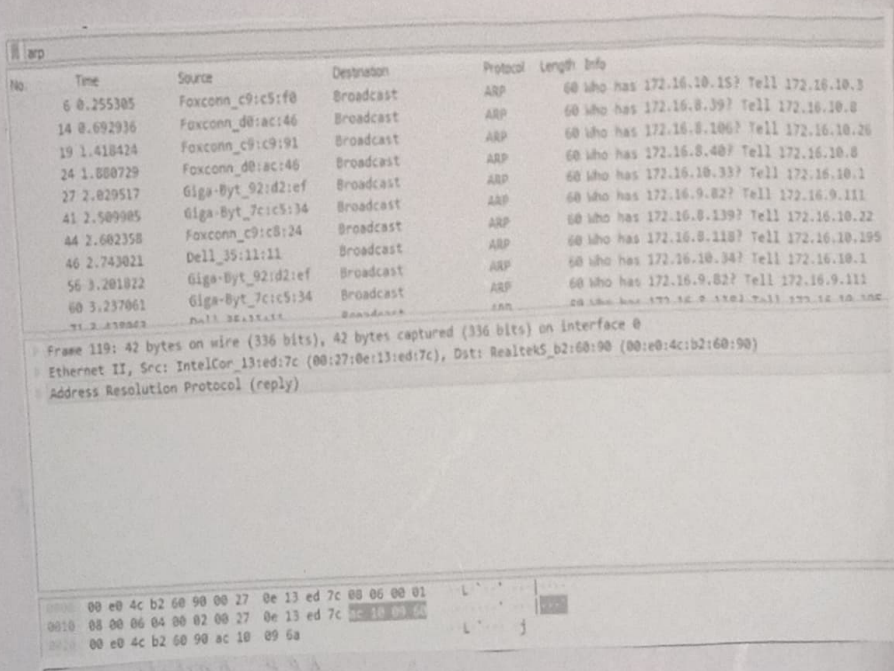
Output:

1. Create a filter only to display TCP packets, inspect the packets.

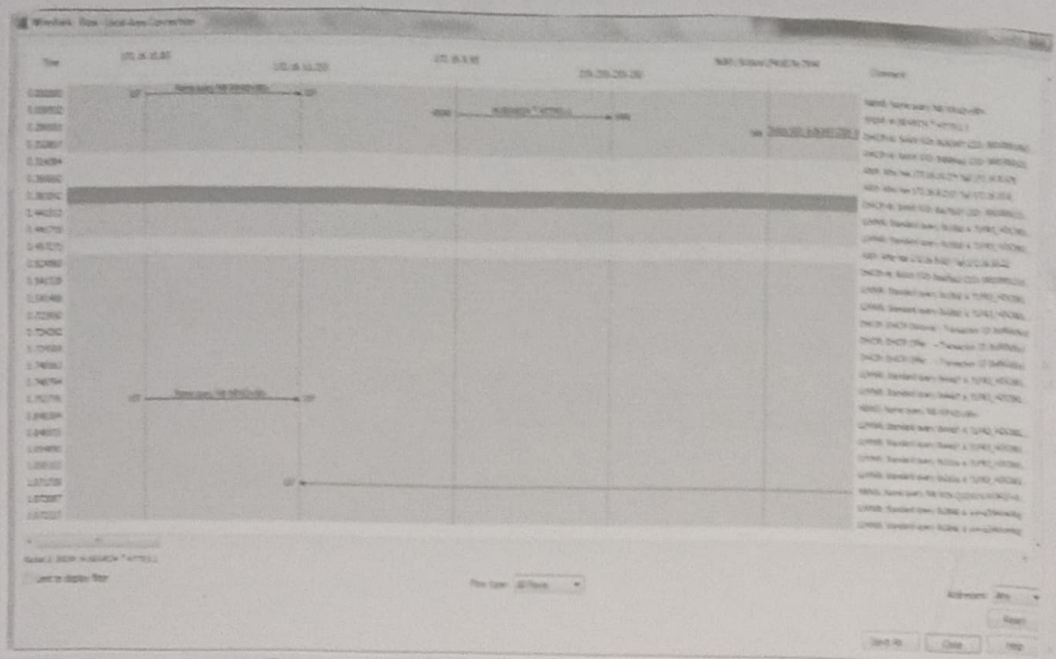
2. Create a filter only to display ARP packets.

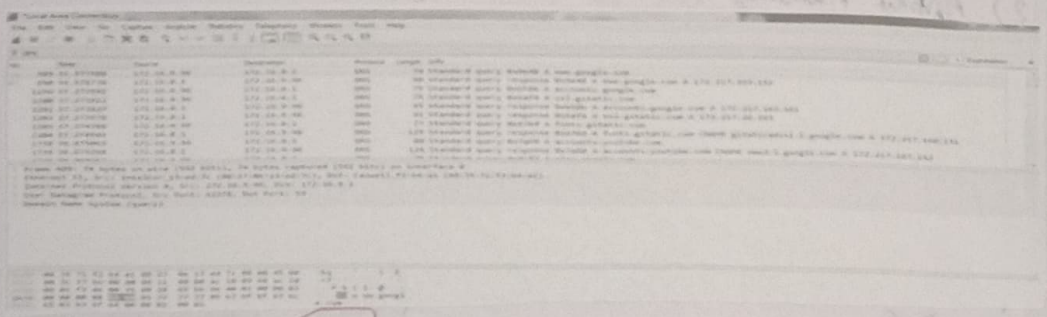


3. Create a filter only to display DNS packets



4. Create a filter to display only DHCP packets





Result :

The experiment on wireshark is completed successfully.

Expt. No: 6 Error Detection - Hamming

Aim

Write a program to implement error detection and correction using hamming code concept.

Procedure:

→ Create a sender program with below features:

1) Input to sender file should be a text.

2) Apply hamming code concept on binary data.

3) Save the output.

→ Create a reciever program with below features.

1) Reciever program should read the input.

2) Apply Hamming code on binary data.

3) If there is error, display position of error.

Program: Sender code

```
def calculate_parity_bits(data_bits):
    r=0
    while(2**r < data_bits + r + 1){
        r+=1;
    }
    return r;
def hamming_encode(data):
    m = len(data)
    r = calculate_parity_bits(m);
    total_bits = m+r
    hamming = [0]*[total_bits+1]
    data_index = 0
```

```
for position in range [1, totalbits+1];
    if position & [position-1] :
    hamming [position] : int (data [data_index])
    data_index += 1
    parity-position = 1
while parity-position <= total bits :
    parity = 0
for position in range (1, total_bits+1) :
    if position & parity-position:
        parity = hamming [position]
encoded = ""
for position in range (1, total_bits+1) :
    encoded += str (hamming [position])
return encoded
with open ('input.txt', 'r') as file :
    text = file.read ();
print ("Original text:")
print (text)
with open ("channel.txt", 'w') as channel:
    for character in text :
        channel.write (encoded + " ");
print ("\n Hamming encoded data saved to
        channel.txt")
```

Reciever code:

```
def hamming_decode(block):
    total_bits = len(block)
    hamming = [0]
for bit in block:
    hamming.append(int(bit))
    error_position = 0
    parity_position = 1
    while parity_position <= total_bits:
        parity = 0
    for position in range(1, total_bits+1):
        if position < parity_position:
            parity ^= hamming[position]
    if position & (position - 1):
        data += str(hamming[position])
    character = chr(int(data, 2))
    return character
with open('channel.txt', 'r') as channel:
    print("Recieved data")
    result = ""
    for block in data:
        character = hamming_decode(block)
        result += character
        print("\n Original message after
                                correction")
        print(result)
```

Result:

The program has been executed successfully.

9/10

Expt No. 7 Flow Control at Data Link Layer

Aim:

Write a program to implement flow control at data link layer using sliding window protocol.

Program:

Sender program:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX 100
struct frame {
    int frame_no;
    char data;
};
int main() {
    struct frame frame[MAX];
    int windowsize, n, i, start=0;
    char message[MAX];
    file fp;
    int ack;
    scanf("%d", &windowsize);
    scanf("%s", &message);
    n = strlen(message);
    for(i=0; i<n; i++) {
        frame[i].frame_no = i+1;
        frame[i].data = message[i];
```

```
while (start <n) {
    int end = start + windowsize;
    if (end > n)
        end = n;
    fp = fopen("sender_Buffer", 'w');
    if (fp == NULL) {
        printf("error");
        return 1;
    }
    for (i = start; i < end; i++) {
        fprintf(fp, "%d %c\n", frames[i].frameno, frames[i].data);
        printf("Frame No: %d DATA : %c \n", frame[i].frame_no,
                                            frame[i].data);
    }
    fclose(fp);
    printf("Frames stored in sender buffer");
    fp = fopen("Recieved Buffer", 'r');
    if (fp == Null) {
        continue;
    }
    fscanf(fp, "%d", &ack);
    fclose(fp);
    printf("Recieved ACK / NACK : %d \n", ack);
    if (ack >= start + 1 && ack <= end) {
        printf("ACK recieved", ack);
        start = ack;
        printf("Sliding window moved forward \n");
    }
}
```

```
else if (ack <= 0) {
    ack = -ack;
printf("NAck recived for frame : %d \n", ack);
start = ack - 1;
}
printf("All frames transmitted successfully \n");
return 0;
}
```

Reciver program :

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main() {
    FILE *fp, *out;
    int frame_no;
    char data;
    int expected = 1;
    while (1) {
        fp = fopen("sender Buffer", "r");
        if (fp == NULL) {
            printf("waiting");
            sleep(2);
            continue;
        }
    }
```

```
while (fscanf (fp, "%d %c", &frame_no, &data) {
    if (frame_no == expected) {
        out = fopen ("Reciever Buffer", "w");
    if (out == null) {
        fclose (fp);
        return 0;
    }
}
fclose (fp);
return 0;
}
}
```

Result:

The code has been executed successfully.

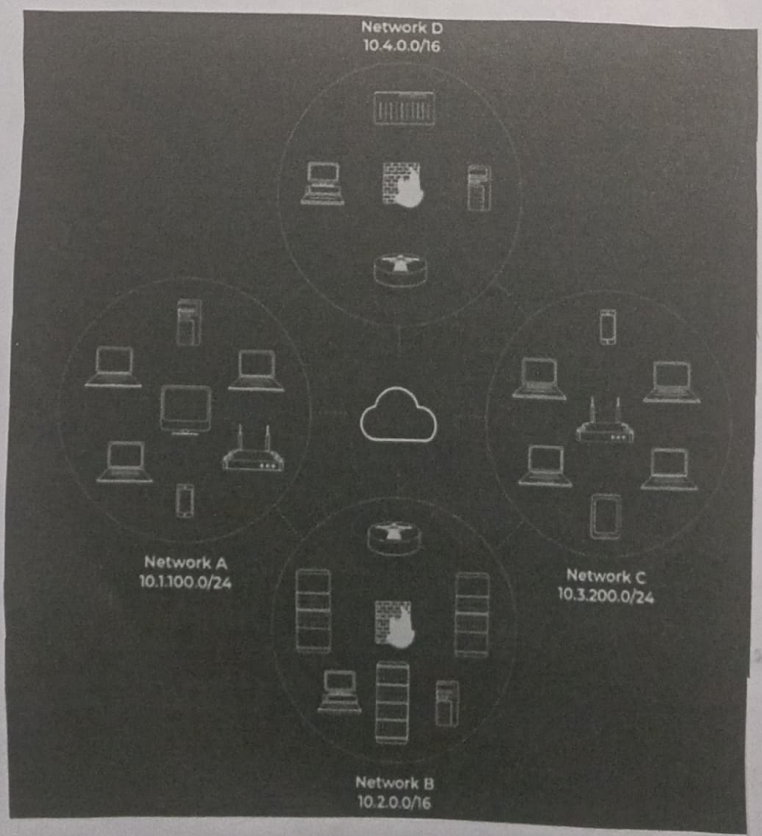
Exp. No. 8 (a) NMAP to discover live hosts

Aim: To discover live host using NMAP scans (ARP, ICMP, TCP/UDP) on hyperhone platform roam link.

Introduction:

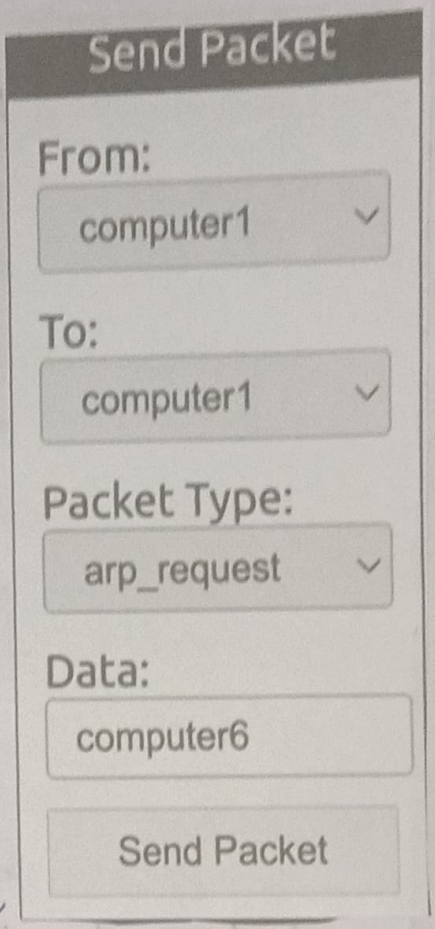
When targeting a network, we need an efficient tool to handle repetitive tasks. First of four in NMAP series:

→ Nmap live host discovery
→ Nmap basic port scans
→ Nmap advanced port scans
→ Nmap Post Port Scans.

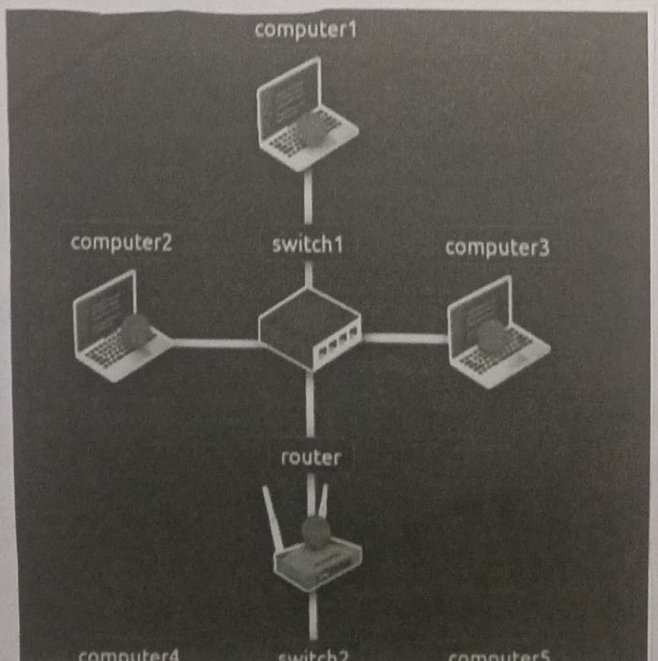


This figure displays 2 types of subnets:

* /16 subnets
* /24 subnets



From computer 1 to computer 6, packet type: 'ARP request'



From computer 4 to computer 6,

Packet type = 'ARP request'

Send Packet

From:

computer4

To:

computer4

Packet Type:

arp_request

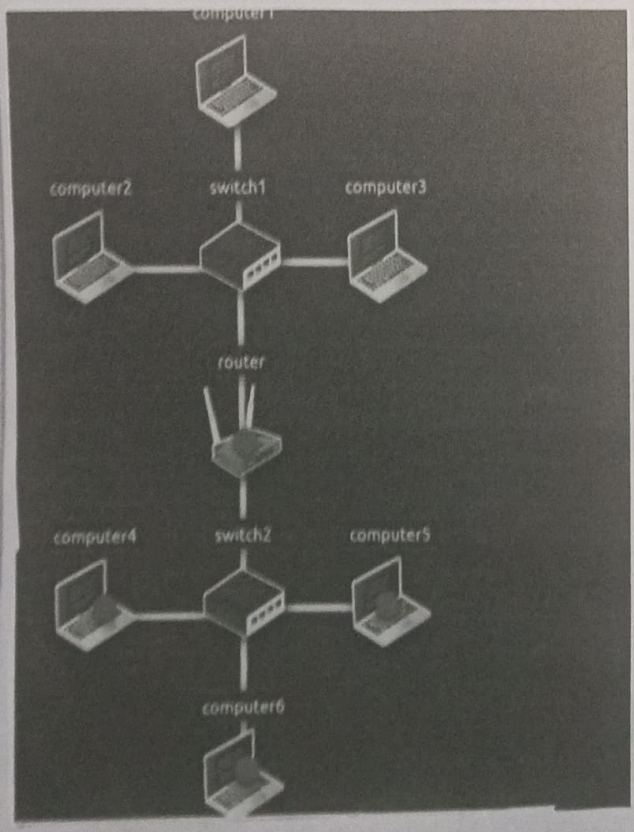
Data:

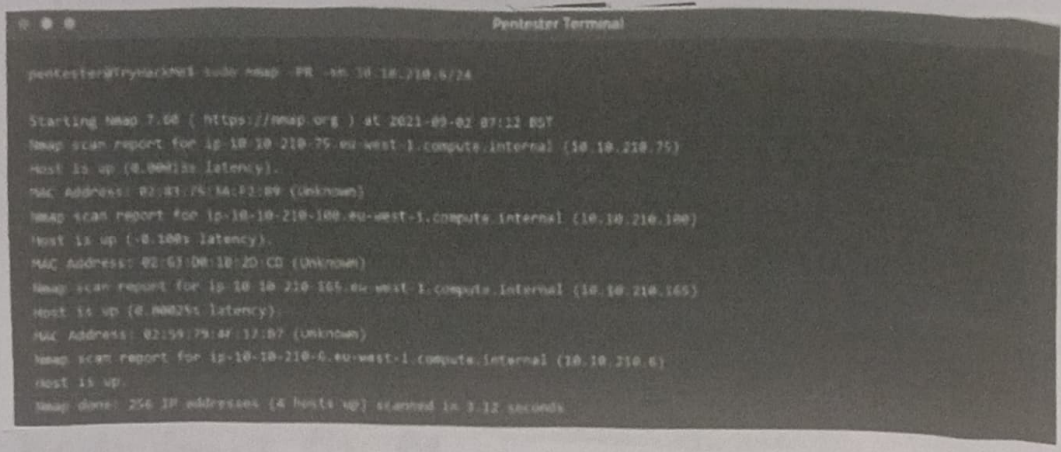
computer6

Send Packet

NMAP host discovery using ARP:

When a privileged user tried to scan targets on local networks, Nmap uses ARP request.





NMAP host discovery using ICMP :

A simple method to identify live hosts on target network is by bringing each IP address and checking for responses.

UDP ping :

While sending UDP packets to open UDP ports may not trigger a response, sending them to closed UDP ports.

Result :

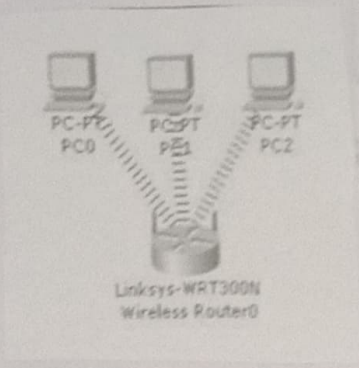
The result has been completed successfully.

Exp. No. 8 (b) Configuration of wireless LAN using CISCO packet tracer.

Aim:

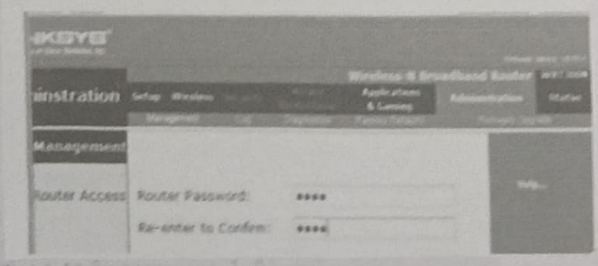
Configuration of wireless LAN using CISCO packet TRACER.

Design a topology with three PCs connected from Linksys wireless routers.



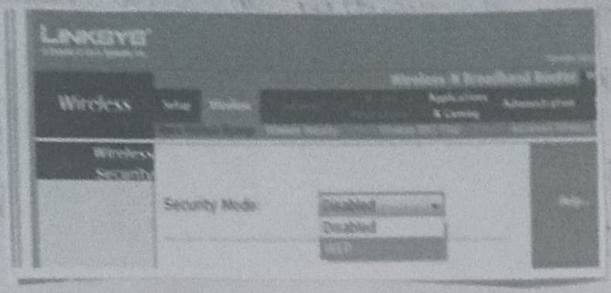
Procedure:

Step-1: Click on wireless routers, select admin tab from top menu.

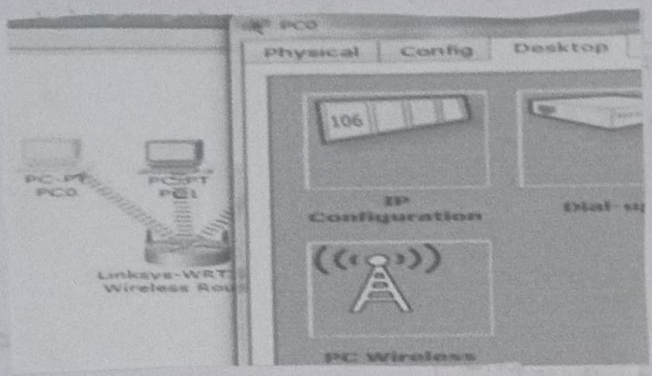


Step-2: Now select wireless security & change security mode to WEP.

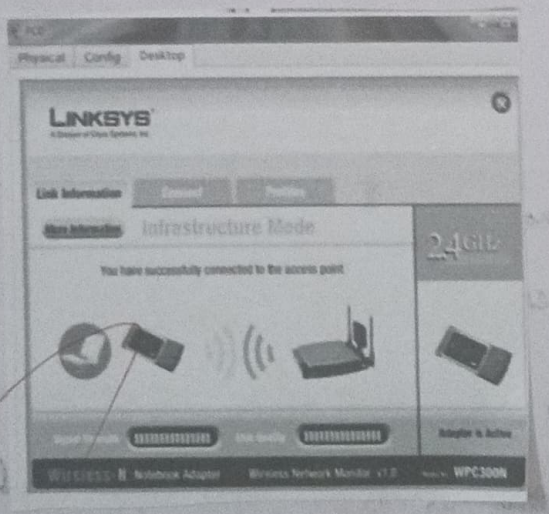
Step 3: Set key 1 to 0123456789



Step-4: Connect PCs from wireless routers



Step-5: Click on connect tab and click on refresh button.



Result:

The configuration has been completed successfully.

8/13 x 9/10

Expt. No. 9 Implementation of subnetting in CISCO packet TRACER simulation.

Aim: Implementation of subnetting in CISCO packet TRACER simulation.

Classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet mask.

Creating a network topology:

To create a network topology in packet tracer, select "New" button in top, then select network & generic.

Adding the device:

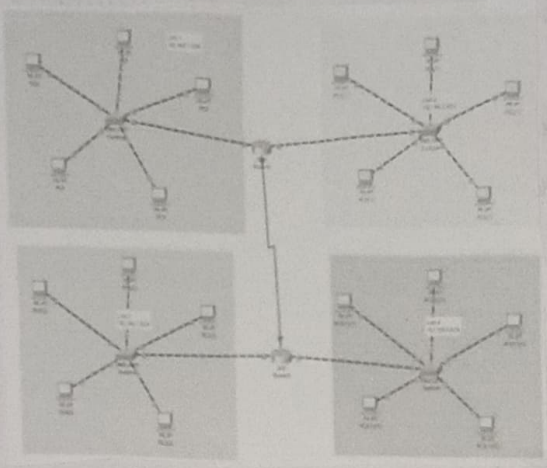
To add a device, select the device from bottom left and drag it onto network topology.

Subnetting:

To subnet the network address of 192.168.10/24 to provide enough space for atleast 5 addresses for end devices, switch and router.

Configuring the devices:

Right click on router and select "CLI" This will open by configuring the router. Right click CLI, enter the following commands:



#enable

#interface FastEthernet 0/0

#ip address [IP address] [subnet mask]

#no shutdown

#exit

Interface Fast ethernet 0/1

#ip address [IP address] [subnet mask]

no shutdown

exit

Result:

The implementation has been completed successfully.

Exp No. 10 (a) Internetworking with router in CISCO Packet TRACER simulator

Aim: Internetworking with router in CISCO Packet TRACER simulator.

Procedure:

Step-1: Configure router 1

Step-2: Configure PCs

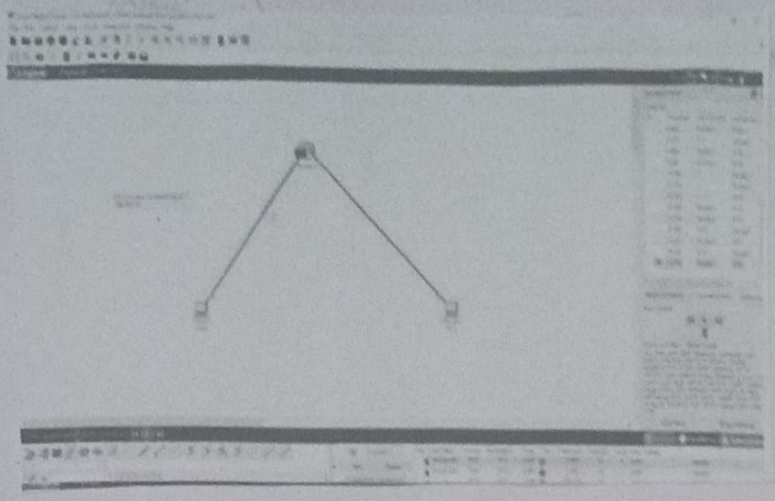
Step-3: Connecting PC with router

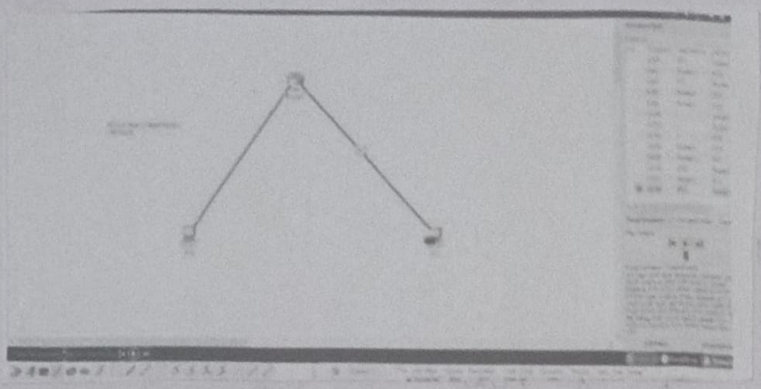
Router Configuration table:

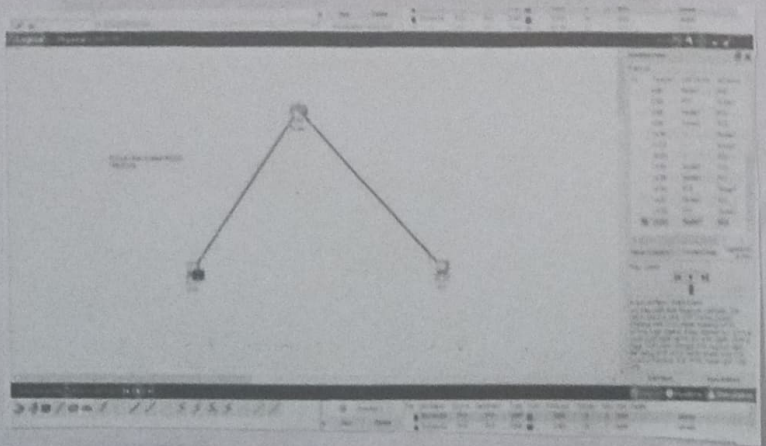
Device name	IP address Ethernet 0/0	Subnet mask	IP address ethernet 0/1	Subnet mask
Router	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0

PC configuration table:

Device name	IP address	Subnet mask	Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1





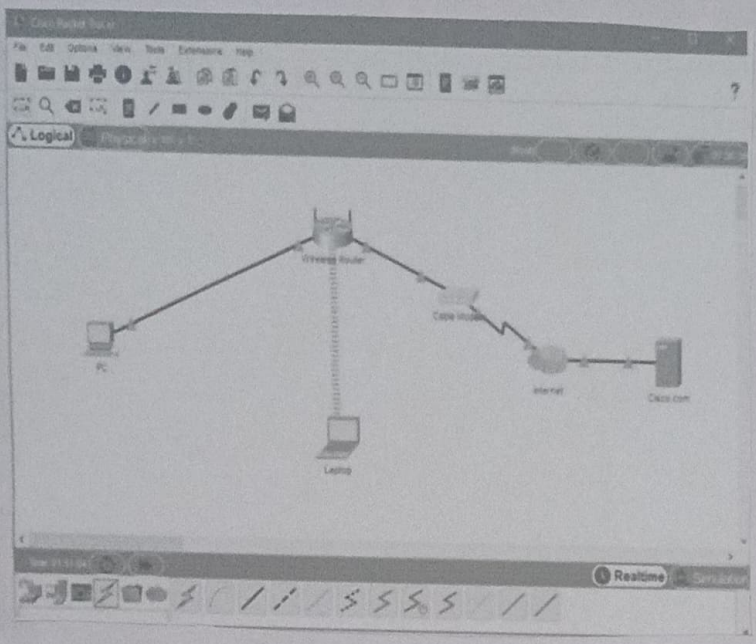


Result: The internetworking with router has been completed successfully.

Exp No: 10 (b) Design and Configure Internetworking

Aim:

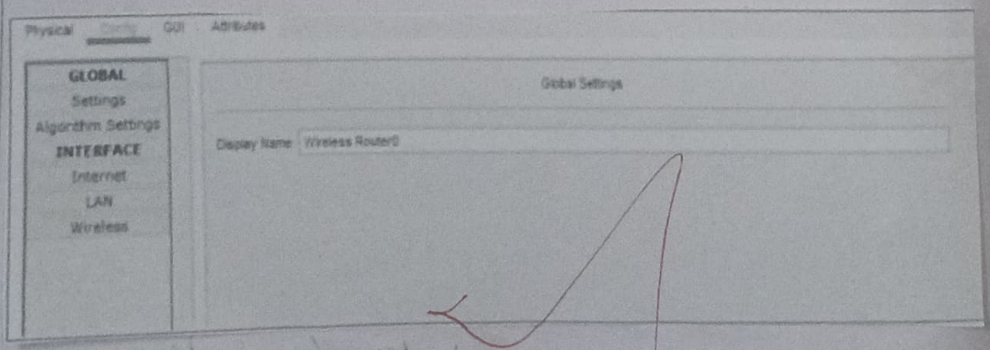
Design and configure an internetworking using wireless router, DHCP server and internet cloud.



Part-1: Build a sample network in the logical topology workspace.

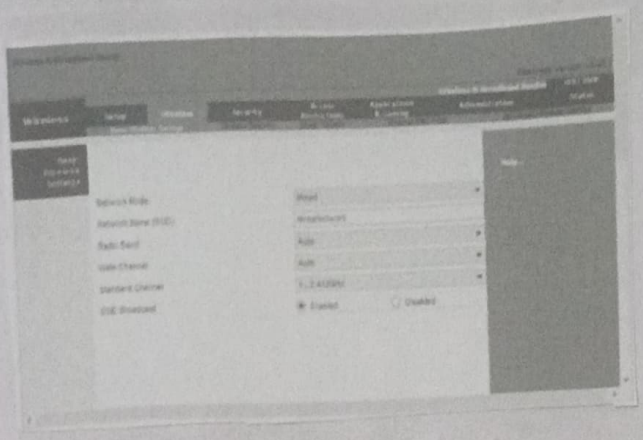
→ Step-1: Launch Packet TRACER

→ Step-2: Build Topology

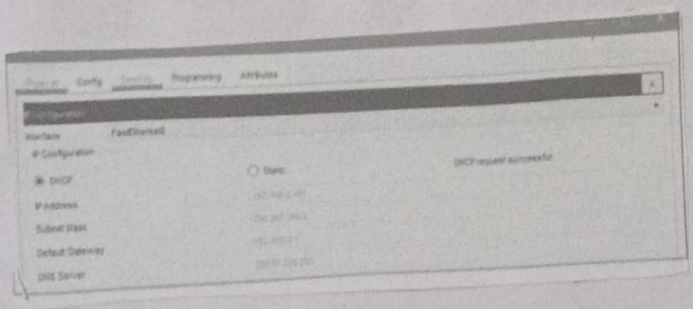


Part-2 - Configure the wireless device

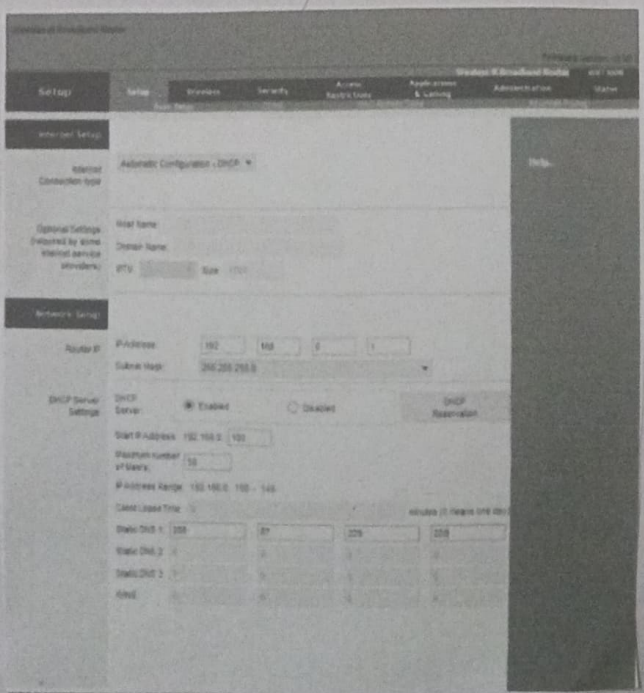
→ Step-1: Configure the wireless router



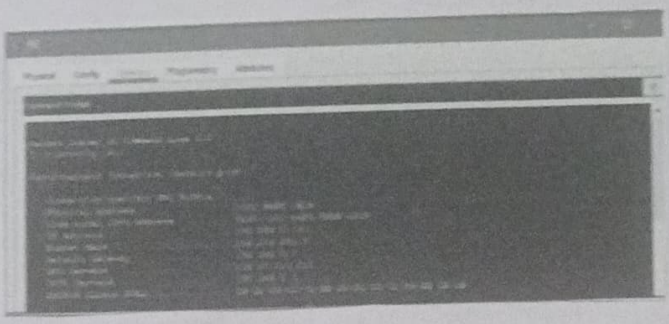
→ Step-2: Configure DHCP server.



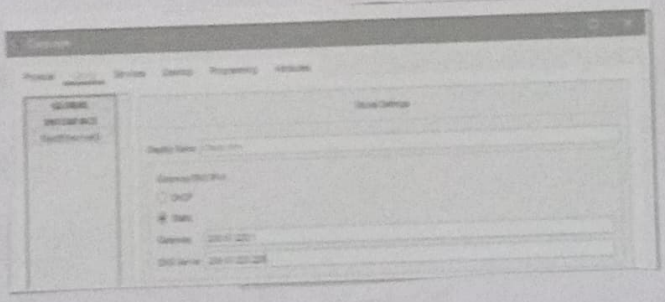
→ Step-3: Configure laptop

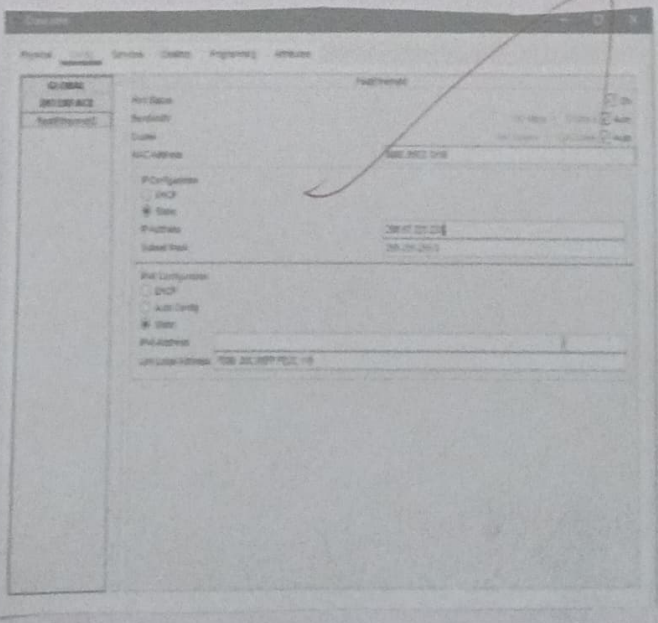


Step-4: Configure the PC

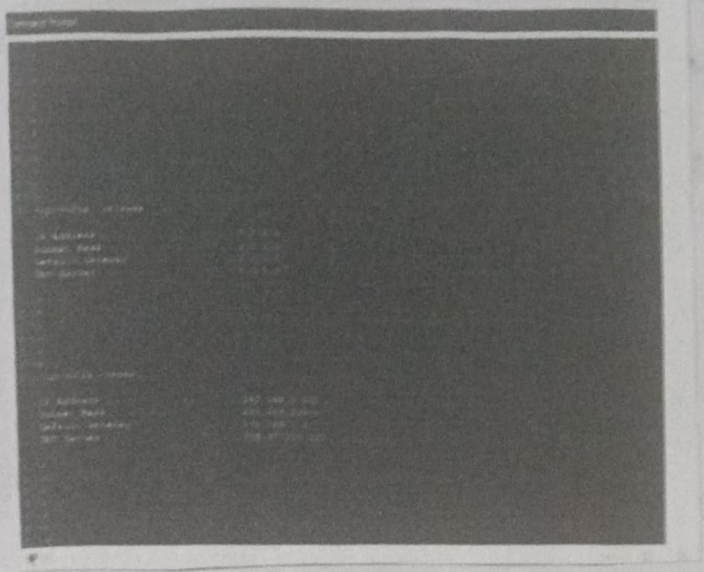


Step-5: Configure interface and





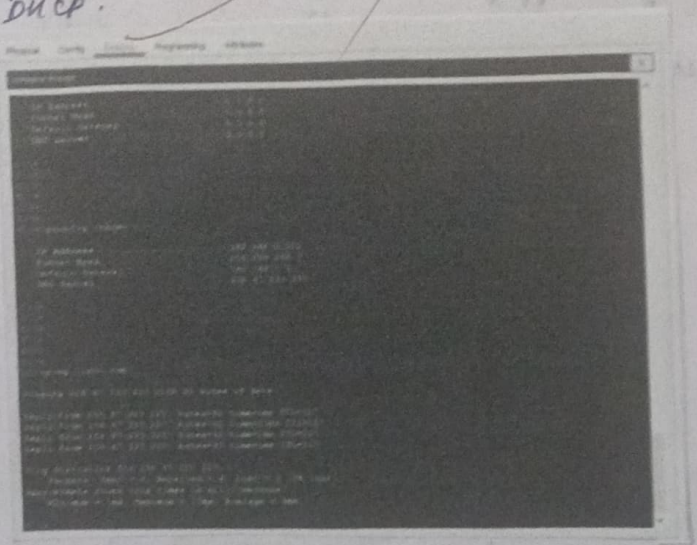
Step-6. Configure the cisco.com server



Part-3: Verify Connectivity

Step-1: Refresh the IPV4 setting on PC.

a) Verify that PC is receiving IPV4 configuration information from DHCP.



b) Test connectivity to cisco.com server from PC.

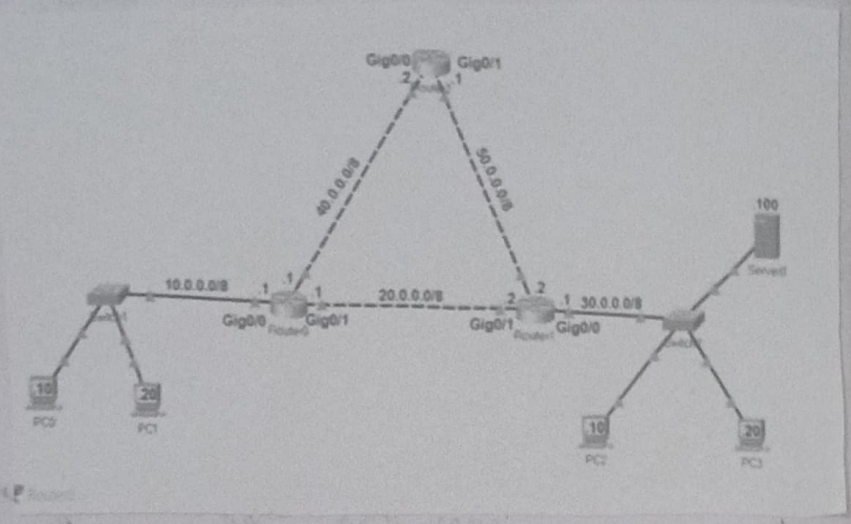
Result -

This experiment has been completed successfully.

Expt. No. 11(a) Simulate Static Routing Configuration

Aim: Simulate static routing configuration using Cisco packet tracer.

Procedure: Setting up a practice lab.



Creating, adding, verifying static routes

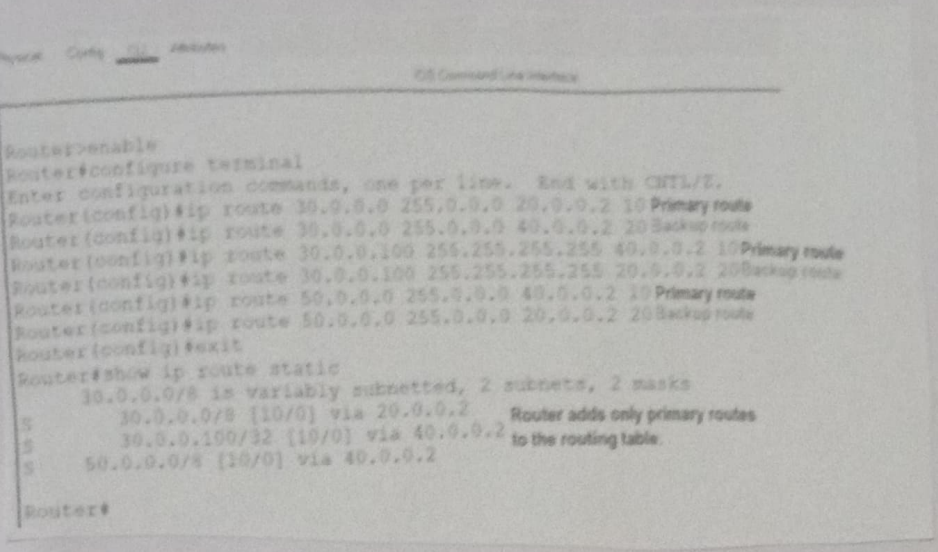
Connected network of each router.

Router	Available on local	Available on other router
Router 0	10.0.0.0/8	30.0.0.0/8
	20.0.0.0/8	50.0.0.0/8
Router 1	40.0.0.0/8	10.0.0.0/8
	30.0.0.0/8	40.0.0.0/8

Router 0 configuration:

Router 0 enable

Router # configure terminal



Router (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.20,

Router (config) # ip route 30.0.0.0 255.0.0.0 4.0.0.30

router (config) # exit

Router 1: configuration

Router 1 enable

Router 1 # configure

terminal

Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.10

Router (config) # ip route 100.0.0.0 255.0.0.0 50.0.0.20

Router (config) # exit

Router2

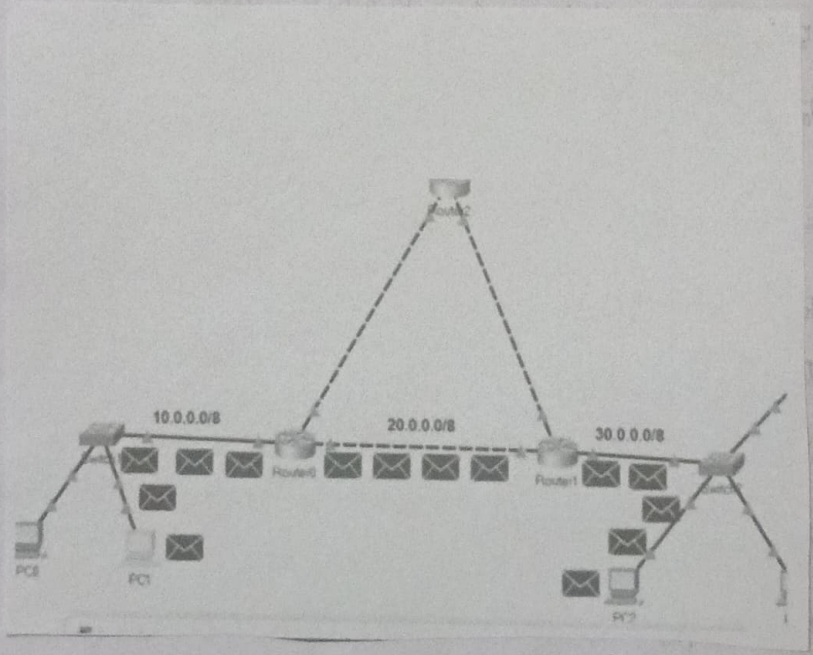
Physical Config CLI Attributes

IOS Command Line Interface

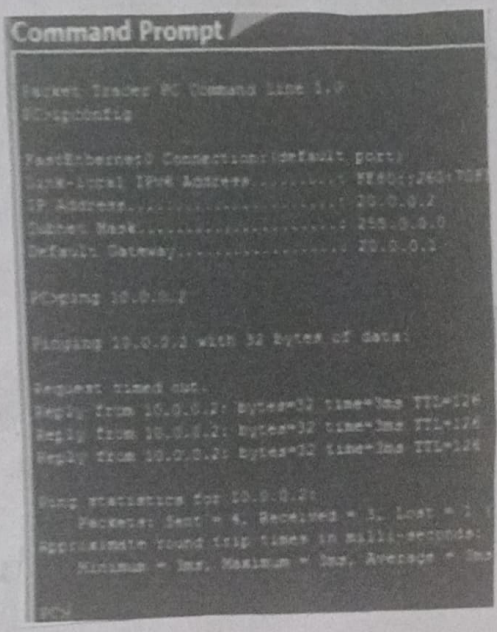
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
S    10.0.0.0/8 [1/0] via 40.0.0.1
S    30.0.0.0/8 [1/0] via 50.0.0.2

Router#
```

Verifying static routing



To simulate the failure of main route you can delete the link between router 0 and router 1



Result:

The experiment has been completed successfully.

9/1/X

Expt. No. 11 (b) Simulate RIP using CISCO packet TRACER

Aim:

Simulate RIP using CISCO packet tracer.

Initial IP configuration:

Device	Interface	IP config	Connected with
PC0	Fast Ethernet	60.0.0.2/8	Router 0's Fa
Router 0	Fa 0/1	10.0.0.1/8	PC0's fast ethernet
Router 0	S0/0/1	192.168.1.254/30	Router 2's S0/0/1
Router 1	S0/0/1	192.168.1.253/30	Router 0's S0/0/0
Router 1	S0/0/1	192.168.1.250/30	Router 0's S0/0/1
PC1	Fast Ethernet	20.0.0.2/30	Router 2's Fa 0/1

Assign IP address to PCs.

Double click PCs and click desktop menu item and click IP config

Router 0:

Router (config)# Interface fast Ethernet 0/0

Router (config)# ip address 10.0.0.1 255.0.0.0 Router (config)# no shutdown

Router (config-if)# exit.

Router 1:

Router -> enable

Router # configure

Router(config) # interface serial 0/0/0

Router (config-if) # no shutdown

Router (config-if) # exit

Router-2:

Router -> enable

Router # configure

terminal

Router (config) # interface fast Ethernet 0/0

Router (config-if) # ip address 20.0.0.1

255.0.0.0 Router (config-if) # no shutdown

Router (config-if) # exit

Access the command prompt of PC1 and use ping command to test connectivity from PC0.

Result:

The experiment has been completed successfully.

Expt. No. 12(a) Implement echo client server using TCP/UDP sockets.

Aim:

Implement echo client server using TCP/UDP sockets.

Algorithm:

TCP server

1. Start the program
2. Create serversocket using a port number
3. Wait using accept();
4. Create I/P & O/P stream
5. Recieve message from client.
6. Send same back to client
7. Stop

TCP - client:

1. Start the program
2. Create a socket & connect to server using port no.
3. Create I/P & O/P streams.
4. Read a message from user.
5. Send same to server
6. Recieve the echoed message
7. Stop.

I/P:

Enter message: Hello server

O/P:

Server started

Waiting for client

client connected

Message from client: Hello server.

Result:

The program has been executed successfully.

13/x

Expt. No. 12 (b)

Implement a chat-client server application

Aim:

To implement a chat-client server application using TCP/UDP sockets.

Algorithm:

TCP chat server:

1. Start the program
2. Create server socket with TCP
3. Wait using accept()
4. Create I/P & O/P streams
5. Recieve message from client
6. Display message
7. Send reply to client
8. Close socket and stop program

TCP chat client:

1. Start the program
2. Create socket & connect the server
3. Create I/P & O/P streams
4. Read a message from client server.
5. Send message from client server.
6. Display reply
7. Close the socket and stop program.

I/P:

Client : Hello

Server : Hi

Client : How are you?

Server : I am fine

Client : Exit

O/P:

Server started

Waiting for client

Client connected

Result:

The program has been executed successfully.

Exp. No. 13 Implement your own ping program

Aim: To write code to implement own ping program.

Program:

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
int main(){
    int sock;
    char buffer[1024];
    struct sockaddr_in addr;
    struct icmphdr icmp;
    sock = socket(AF_INET, SOCK_RAW, IPPROTO_ICMP);
    if(sock<0){
        perror("socket");
        return 1;
    }
    memset(&icmp, 0, sizeof(icmp));
    icmp.type = ICMP_ECHO;
    icmp.code = 0;
    sendto(sock, &ICMP, sizeof(icmp), 0, (struct sockaddr*)
           &addr, sizeof(addr));
    printf("Reply received from 8.8.8.8\n");
    close(sock);
    return 0;
}
```

Result:

The program has been executed successfully.

Exp. No 14 Implement Packet Sniffing using Raw sockets

Aim:
Write a code using Raw sockets to implement packet sniffing.

Program:

```
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>
int main(){
    int sock;
    char buffer[65536];
    struct sockaddr saddr;
    int len = sizeof(saddr);
    while(1){
        int n = recvfrom(sock, buffer, sizeof(buffer), &saddr,
                         &len);
        if(n>0){
            struct iphdr *ip = (struct iphdr *) buffer;
            printf("Packet recieved\n");
        }
        close(sock);
        return 0;
}
```

Result:
The program has been executed successfully.

Exp. No-15 Analyse various types of servers using WebAlizer tool

Aim:

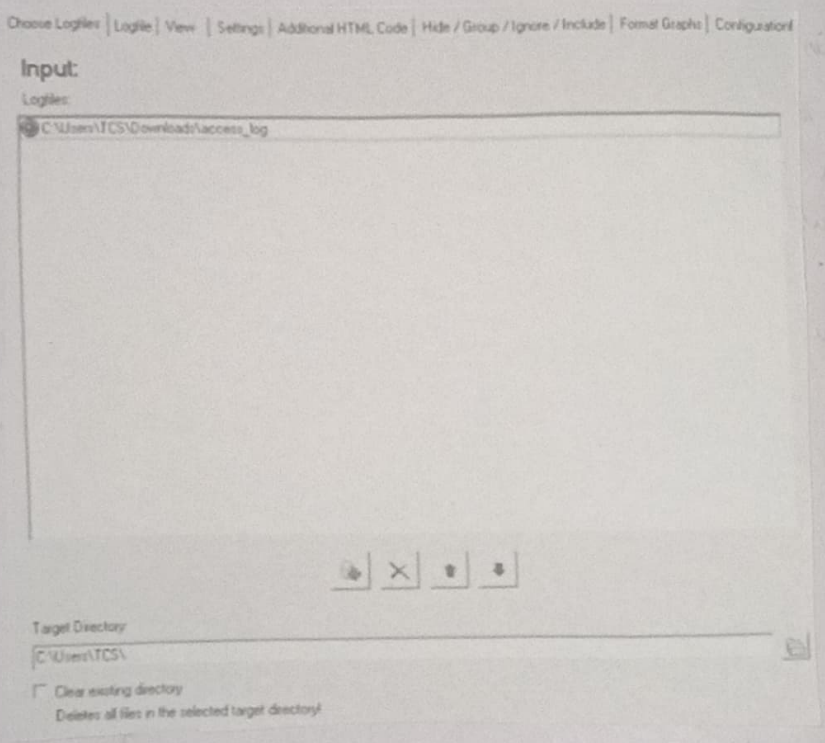
To analyse the different types of web logs using the webalizer tool.

Procedure:

Step-1: Run webalizer windows version

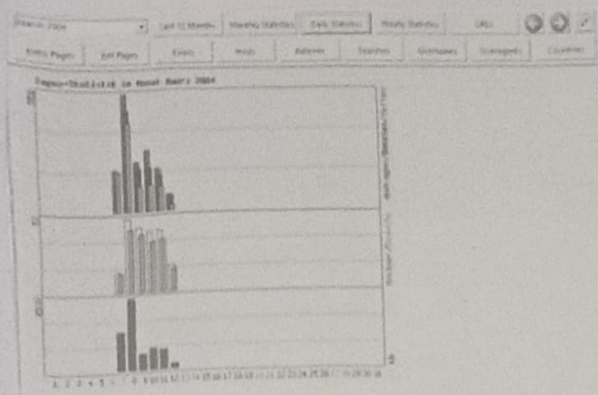
Step-2: Input web log file

Step-3: Press run webalizer



Output.

Monthly station:

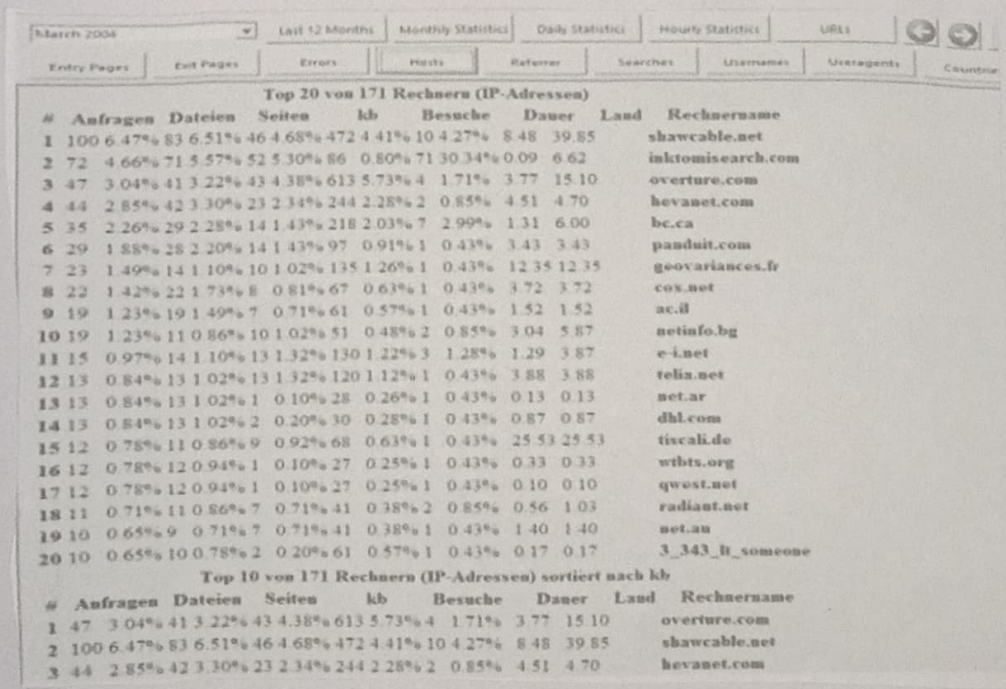


Seiten - Diskutierte Abfragen (z.B. html, asp, usw). Dateien - erfolgreiche Anfragen für alle Typen (z.B. html, gif, usw). Treffer - alle Anfragen, inklusive Fehler.

Tages-Statistik im Monat Maerz 2004

Tag	Anfragen			Dateien			Seiten			Besuche			Rechner			kb		
	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum	Summe	Schnitt	Maximum
7	197 12.74%	25	29	147 11.56%	18	21	116 18.04%	21	27	15 6.41%	2	3	14 8.19%	2	3	2383 22.27%	298	542
8	459 16.16%	24	61	478 17.42%	21	56	419 42.67%	18	40	97 24.38%	2	7	48 28.07%	2	7	4509 42.14%	186	886
9	232 15.01%	11	29	205 16.09%	10	29	118 12.02%	6	16	49 20.96%	2	5	44 25.73%	2	5	955 8.95%	45	138
10	288 18.56%	14	102	295 16.09%	16	34	124 12.63%	6	28	46 19.66%	2	8	38 22.22%	2	4	1136 12.49%	67	376

Hosts:



Top 20 von 171 Rechnern (IP-Adressen)

#	Anfragen		Dateien		Seiten		kb		Besuche		Dauer		Land	Rechnername
1	100	6.47%	83	6.51%	46	4.68%	472	4.41%	10	4.27%	8.48	39.85		shawcable.net
2	72	4.66%	71	5.57%	52	5.30%	86	0.80%	71	30.34%	0.09	6.62		inktomisearch.com
3	47	3.04%	41	3.22%	43	4.38%	613	5.73%	4	1.71%	3.77	15.10		overture.com
4	44	2.85%	42	3.30%	23	2.34%	244	2.28%	2	0.85%	4.51	4.70		hevanet.com
5	35	2.26%	29	2.28%	14	1.43%	218	2.03%	7	2.99%	1.31	6.00		bc.ca
6	29	1.88%	28	2.20%	14	1.43%	97	0.91%	1	0.43%	3.43	3.43		panduit.com
7	23	1.49%	14	1.10%	10	1.02%	135	1.26%	1	0.43%	12.35	12.35		geovariances.fr
8	22	1.42%	22	1.73%	8	0.81%	67	0.63%	1	0.43%	3.72	3.72		cos.net
9	19	1.23%	19	1.49%	7	0.71%	61	0.57%	1	0.43%	1.52	1.52		ac.il
10	19	1.23%	11	0.86%	10	1.02%	51	0.48%	2	0.85%	3.04	5.87		netinfo.bg
11	15	0.97%	14	1.10%	13	1.32%	130	1.22%	3	1.28%	1.29	3.87		e-i.net
12	13	0.84%	13	1.02%	13	1.32%	120	1.12%	1	0.43%	3.88	3.88		telia.net
13	13	0.84%	13	1.02%	1	0.10%	28	0.26%	1	0.43%	0.13	0.13		net.ar
14	13	0.84%	13	1.02%	2	0.20%	30	0.28%	1	0.43%	0.87	0.87		dhl.com
15	12	0.78%	11	0.86%	9	0.92%	68	0.63%	1	0.43%	25.53	25.53		tiscali.de
16	12	0.78%	12	0.94%	1	0.10%	27	0.25%	1	0.43%	0.33	0.33		wtbts.org
17	12	0.78%	12	0.94%	1	0.10%	27	0.25%	1	0.43%	0.10	0.10		qwest.net
18	11	0.71%	11	0.86%	7	0.71%	41	0.38%	2	0.85%	0.56	1.03		radiant.net
19	10	0.65%	9	0.71%	7	0.71%	41	0.38%	1	0.43%	1.40	1.40		net.au
20	10	0.65%	10	0.78%	2	0.20%	61	0.57%	1	0.43%	0.17	0.17		3_343_ll_someone

Top 10 von 171 Rechnern (IP-Adressen) sortiert nach kb

#	Anfragen		Dateien		Seiten		kb		Besuche		Dauer		Land	Rechnername
1	47	3.04%	41	3.22%	43	4.38%	613	5.73%	4	1.71%	3.77	15.10		overture.com
2	100	6.47%	83	6.51%	46	4.68%	472	4.41%	10	4.27%	8.48	39.85		shawcable.net
3	44	2.85%	42	3.30%	23	2.34%	244	2.28%	2	0.85%	4.51	4.70		hevanet.com

Result:

The experiment has been executed successfully.